TU/CSE

Tezpur University

Autumn Mid-Semester Examination-2025

Course Code: CSMT537 **Course Title: Recommender Systems**

Full marks=30 *Time= 90 Minutes*

Instructions: There are a total of 6 Questions. Missing data, if any, may be suitably assumed.

1. A supermarket wants to analyze the purchasing behaviour of its customers using association rule mining. The goal is to find patterns among frequently bought items to improve product recommendation and design promotional bundles. The following table (**Table 1**) shows 10 transactions recorded in the store:

Transaction ID	Items Purchased
T1	Milk, Bread, Butter, Eggs, Jam
T2	Bread, Butter, Coffee
T3	Milk, Bread, Butter, Jam
T4	Milk, Eggs, Coffee
T5	Bread, Butter, Eggs, Coffee
T6	Milk, Bread, Jam
T7	Milk, Bread, Butter, Eggs, Coffee
T8	Bread, Butter, Jam, Coffee
T9	Milk, Bread, Butter
T10	Bread, Butter

Table 1: List of household items

a) Identify all frequent itemsets that satisfy the minimum support threshold using Frequent Pattern Growth Algorithm. 30 % [3]

b) Apply Frequent Pattern Growth Algorithm to generate association rules from these frequent itemsets that meet the confidence threshold. 40% [4]

c) Interpret at least two strong rules and explain how the grocery store could use these insights for business decisions (e.g., cross-selling or promotional offers). [2]

d) Calculate the support for the itemset: {Bread, Butter, Coffee} 30 [1]

e) Compute the confidence for the association rule: {Milk, Bread} → {Butter} 50 [1]

f) Determine the lift for the rule: {Bread, Butter} → {Coffee} 1.6 [1]

2. Construct an AVL Tree by inserting the following sequence of keys into an empty tree:
[30, 20, 40, 10, 25, 35, 50, 5]
After each insertion:

a) Compute the balance factor of affected nodes. Then identify where rotations occur and specify the type of rotation (LL, RR, LR, or RL). Draw the final balanced AVL Tree. [3]

b) State the worst-case time complexity for insertion and search operations in an AVL Tree. [1]

c) What is the maximum height of any AVL tree with 47 nodes? [2]

3. Insert the following keys 14, 18, 25, 34, 7, 3, 15, 96 and 11 (strictly in this order) into an initially empty B-Tree of order 3 (m=3). Illustrates the steps of insertion of each element. [3]

4. A complete binary min-heap is constructed by inserting each integer from [1. 2047] exactly once. The depth of a node in the heap is defined as the number of edges in the path from the root to that node (the root being at depth 0). Then determine the maximum possible depth at which the integer 25 can appear in this heap Justify your answer. [3]

5. Consider a binary min-heap containing 107 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. What is the number of possible values of k? [3]

6. Consider B+ tree in which the search key is 14 bytes long, block size is 2048 bytes. record pointer is 8 bytes long and block pointer is 10 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is. Show all intermediate steps. [3]

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Full marks=30 *Time= 90 Minutes*

Instructions: There are a total of 4 Questions, and you need to attempt only 3 Questions. Missing data, if any, may be suitably assumed.

1. Consider a transaction dataset **(Table-1)** consisting of commonly used grocery and kitchen items purchased by customers in a supermarket. The dataset contains 8 transactions, where each transaction lists the items bought together during a single purchase. Assume the minimum support threshold is 50% and the minimum confidence threshold is 50%.

Table-1: Transaction Database

Transaction ID	Items Purchased
T1	Rice, Cooking Oil, Sugar
T2	Rice, Pulses, Cooking Oil, Sugar
T3	Sugar, Tea, Milk
T4	Rice, Sugar, Milk
T5	Pulses, Cooking Oil, Tea
T6	Rice, Pulses, Sugar
T7	Milk, Tea, Sugar
T8	Rice, Cooking Oil, Milk, Sugar

a) Apply the Apriori algorithm to identify all frequent item sets from the given transaction database. Show the candidate generation and pruning steps involved in each iteration of the Apriori algorithm. **[4 M]**

b) Generate all possible association rules from the discovered frequent item sets. Compute the support and confidence of each generated rule. **[4 M]**

c) If a customer has purchased Rice and Sugar, recommend additional items using the strong association rules and justify your recommendation. **[2 M]**

2. Consider the following textual documents consisting of Bollywood movie titles, where each document contains a short phrase. The goal is to cluster the documents based on word similarity.

D1: Dil
D2: Dil Chahta Hai
D3: Dil To Pagal Hai
D4: Dil Hai Tumhaara

Assume that all text is converted to lowercase. Jaccard distance is used as the distance metric.

$$\text{Jaccard Distance} = 1 - \frac{|A \cap B|}{|A \cup B|}$$

a) Represent the documents using binary term vectors. Compute the pairwise Jaccard distance matrix for all document pairs. **[3 M]**

b) Apply the Single Linkage Hierarchical Clustering algorithm to cluster the documents. Show each step of cluster formation, indicating which documents/clusters are merged at every iteration. Draw the dendrogram representing the single linkage clustering process. **[7 M]**

3. Consider the following labelled and noise-free dataset shown in **Table-2**, where each user is described using two attributes: Age and Adventure Preference Level. The decision attribute is the Type of Adventure Activity Recommended. Adventure Preference Level is encoded as: 0 = Low, 1 = Medium, 2 = High.

Table-2: Adventure Activity Booking Transactions

Name	Age	Adventure Preference Level	Recommended Activity
Aman	18	2	Trekking
Neha	25	1	River Rafting
Rohit	35	0	Camping
Pooja	22	2	Trekking
Kavya	30	1	River Rafting
Arjun	20	2	Trekking
Rahul	28	1	River Rafting

a) Apply the K-Nearest Neighbor (KNN) algorithm to predict type of adventurous activity that should be recommended to a new user Ritika, who is 16 years old and has a high adventure preference (Preference Level = 2). Use Euclidean distance for nearest-neighbor calculation and assume K = 3. **[7 M]**

b) A music streaming application collects data such as song likes, listening duration, skipped tracks, and user reviews. Classify these data sources as explicit or implicit feedback. **[3 M]**

4. Consider the following user–item utility matrix (**Table-3**) with 7 users, who have rated three movies. The ratings are on a scale of 1 to 5, indicating the users' likability of the movies. The symbol '–' indicates that the user has not rated the movie. Assume k = 2 nearest neighbors when computing the predicted ratings.

Table-3: User-item utility matrix

User	Pathaan	Tiger 3	Jawan
U1	5	4	5
U2	4	5	4
U3	5	5	5
U4	3	2	3
U5	4	4	4
U6	2	3	2
U7 (Target)	4	3	–

$\frac{.33 \times 4 + .33 \times 3}{.33 + .33}$

a) Calculate the predicted rating of Movie "Jawan" for user U7 using user-based collaborative filtering technique. **[8 M]**

b) Does the user-based collaborative filtering approach suffer from the cold start problem? Justify your answer. **[2 M]**